[METRIC]
A-A-52516
August 14, 1995
SUPERSEDING
MIL-P-62424(AT)
16 August 1991
(see 5.5)

COMMERCIAL ITEM DESCRIPTION

POWDER, YTTRIUM, ALUMINUM, NICKEL-CHROMIUM (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers nickel-chromium, aluminum, yttrium (94Ni-Cr, 6Al, 1.0Y) powder used in plasma applications. This powder is primarily intended for use as a plasma spray deposited material providing increased oxidation and hot corrosion resistance.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's materials specification. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 5.5).

2.1.1 Process. Powder shall be produced by vacuum or inert atmosphere melting process followed by atomizing in an inert atmosphere.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E, Warren, MI 48397-5000.

FSC 9630

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2.1.2 Particle size. The particle size of the powder shall be as specified in table I.

TABLE I. Particle size distribution by weight.

Sieve designation 1/		Percentage (%) by weight (Maximum)	Weight passed through sieve	Weight Retained on sieve
Micrometer	Mesh			
75	200	5		X
45	325	20	X	

1/ Sieve designation in accordance with ASTM E11.

2.1.3 Chemical composition. The chemical composition of the material shall conform to the percent by weight for each element as specified in table II.

TABLE II. Chemical composition.

Elements	(%) by weight	
	Minimum	Maximum
Carbon	---	0.02
Chromium	21.00	23.00
Aluminum	9.00	11.00
Yttrium	0.80	1.20
Phosphorus	---	0.010
Sulfur	---	0.010
Nitrogen 1/	---	0.020 (200 ppm) 2/
Hydrogen 1/	---	0.005 (50 ppm)
Other elements, total 3/	---	0.50
Nickel	Remainder	

1/ Determination of nitrogen and hydrogen elements shall not be required for acceptance.

2/ ppm = parts per million.

3/ Other elements are the sum percentage of individual elements that may be present but not mentioned specifically in the composition. The producer normally shall analyze only for impurities (elements) which are possibly present because of raw materials or manufacturing processes and which affect the product significantly. It is not implied by this statement that an analysis need be made for each element of the periodic table not mentioned specifically in the composition.

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2.1.4 Properties. The powder shall meet the following properties requirements when applied (plasma sprayed) to steel and cobalt, and nickel-base alloys.

a. Bond strength The bond strength of the plasma-sprayed material shall be not less than 28 megapascals (MPa).

b. Cup test. The amount of draw before spalling of the coating shall not be less than 7.6 millimeters (mm).

2.2 Unit containers. Unless otherwise specified (see 5.3), the unit containers shall be capable of holding the following quantities of powder: 0.5, 2.5 and 10 kilograms (Kg). The containers shall be sealed moisture proof units and large enough to permit mixing the powder within before use. The containers shall be capable of protecting the powder against damage or contamination until it is ready to be used.

2.3 Identification markings. The identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, part number, lot number and the part or identification number (PIN) (see 5.3).

2.3.1 Labeling. Unit containers and packages shall be conspicuously labeled "CAUTION: Containers contain powder aluminum. Use approved respirator."

2.3.2 Container. Each container shall be marked "Nickel-chromium, aluminum, yttrium powder".

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

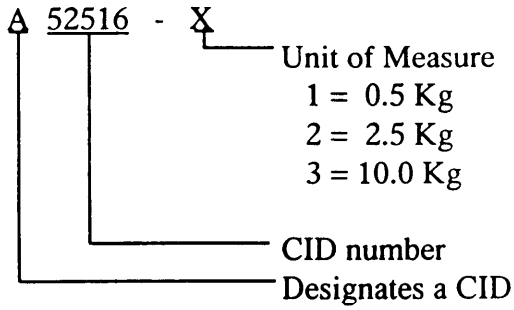
5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. Copies of ASTM E11 "Standard Specifications for Wire Cloth Sieves for Testing Purposes" are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Selection of applicable level and packaging requirements.
d. Unit container and PIN number.
e. Unit container size if other than specified.

5.3 Part or identification number (PIN). The PINs to be used for powder acquired to this CID, are created as follows:



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5.4 Cross reference data. Powder conforming to this CID is interchangeable/substitutable with powder conforming to MIL-P-62424(AT). Cross reference to former part number is as indicated in table IV.

TABLE IV. Cross reference.

Unit Container size	PIN	Former PIN No.
0.5 Kg	A52516-1	MS62424-0.5Kg
2.5 Kg	A52516-2	MS62424-2.5Kg
10.0 Kg	A52516-3	MS62424-10.0Kg

5.5 Supersession data. This CID supersedes Textron Lycoming (Allied Signal) internal specification M3960B, dated 26 April 1982.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AT

(Project 9630-0034)

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